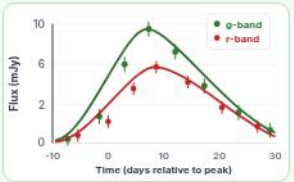
RTF: Real-Time Fingerprints

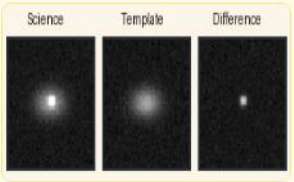
MULTIMODAL INPUTS | INPUT ENCODERS | ENCODER & FUSION | BOTTLENECK | DECODER STACK | RECONSTRUCTION

Alert Photometry & Metadata



$x \in \mathbb{R}^{(B \times L \times 37)}$

Image Cutouts



$I \in \mathbb{R}^{(B \times 3 \times 63 \times 63)}$

Input Projection + Time2Vec
Linear(37 → 256)
+ Periodic time encoding

tokens

[CLS] token
Transformer Encoder
4 layers, 8 heads, Pre-LN
d = 256
$h_CLS \in \mathbb{R}^{(B \times 256)}$

h_CLS

Image Tower
CNN / Zoobot / DINOv2
$e_img \in \mathbb{R}^{(B \times 256)}$

e_img (B, 256)

Additive Fusion
h_CLS + e_img

Latent Vector z
$z \in \mathbb{R}^{(B \times 64)}$
or $\mathbb{R}^{(B \times 128)}$

Broadcast Projection
+ Pos. Embeddings
$z \to \mathbb{R}^{(B \times L \times 256)}$

Transformer Decoder
2 layers, 8 heads, Pre-LN
d = 256

Continuous Head
dt, dt_prev, flux, flux_err
Loss: MSE

Band Head
g, r, i filters
Loss: Cross-Entropy